

Fig. 2: An actual-size, single-side PCB for the water-tank-filling guard system

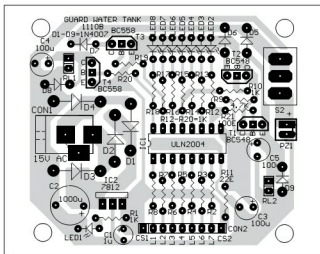


Fig. 3: Component layout for the PCB

PARTS LIST

Semiconductors:

- IC1 - ULN2004 seven-transistor Darlington array
 IC2 - 7812 12V regulator
 T1, T2 - BC548 npn transistor
 T3, T4 - BC558 pnp transistor
 D1-D9 - 1N4007 rectifier diode
 LED1-LED8 - 5mm light-emitting diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R10 - 1-kilo-ohm
 R12-R20 - 47-kilo-ohm
 R2-R8 - 4.7-kilo-ohm
 R9 - 22-ohm
 R11 - 100-ohm

Capacitors:

- C1 - 1 μ F, 16V electrolytic
 C2 - 1000 μ F, 25V electrolytic
 C3-C5 - 100 μ F, 16V electrolytic

Miscellaneous:

- X1 - 230V primary to 15V, 500mA secondary transformer
 S1-S3 - On/off toggle switch
 RL1, RL2 - 12V, 1C/O (single-changeover) relay
 PZ1 - Piezobuzzer
 M1 - 230V, 50Hz, 0.5HP AC motor
 CS1, CS2, L1-L7 - Metallic-strip water-level sensors

L4 through L1, respectively, because of the conductivity of water, creating voltages on these terminals. These voltages are fed, via resistors, to pins 4 to 1 of IC1, respectively. Corresponding output pins of IC1 (pins 13 through 16) go low. The low outputs of IC1 are fed to the respective light-emitting diodes (LED5 through LED8)

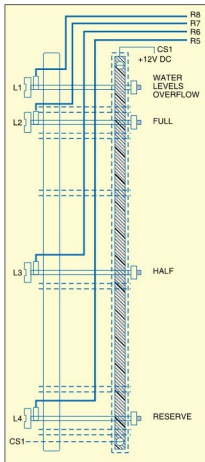


Fig. 4: Overhead tank sensor assembly

to indicate the different levels of water in the overhead tank.

When water in the overhead tank rises to bridge terminals CS1 and L2,

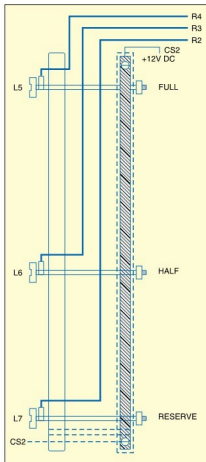


Fig. 5: Ground-tank sensor assembly

input pin 2 of IC1 goes high and IC1 outputs low at pin 15. The base of transistor T3 also goes low through resistor R19, providing 12V to piezobuzzer PZ1